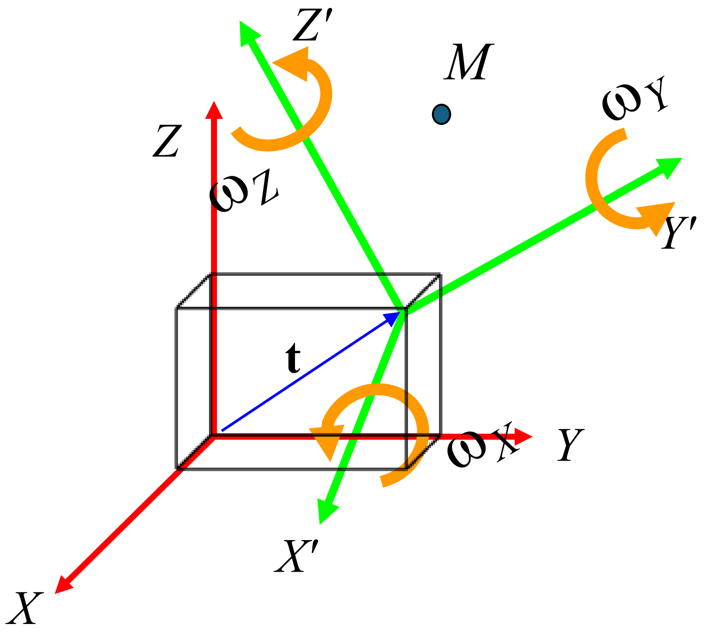
Transformaciones 3D

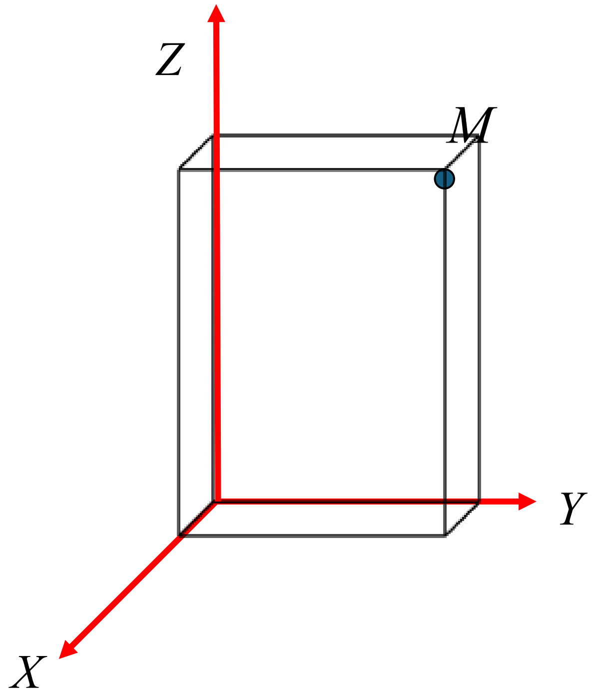
(CAPÍTULO 2 – CV02C)



Domingo Mery

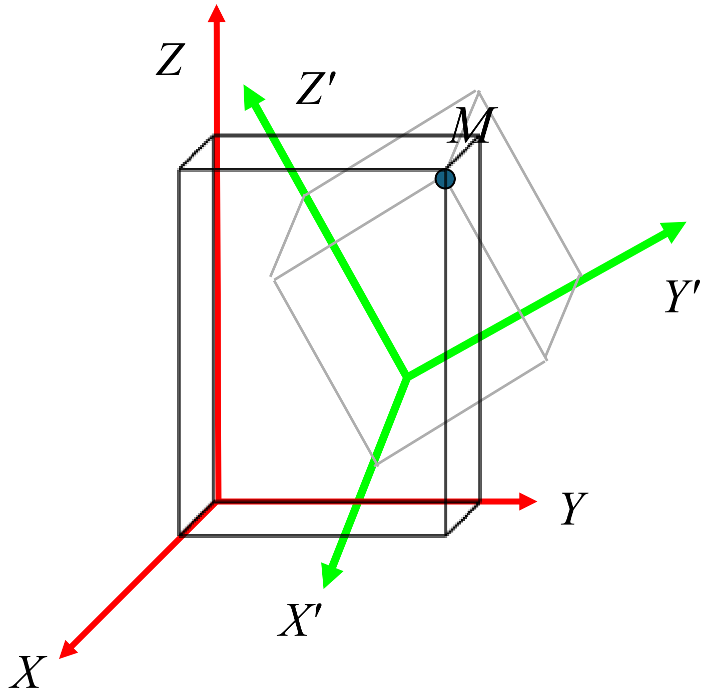
Ciencia de la Computación
Universidad Católica – Chile
http://domingomery.ing.uc.cl

Transformaciones 3D



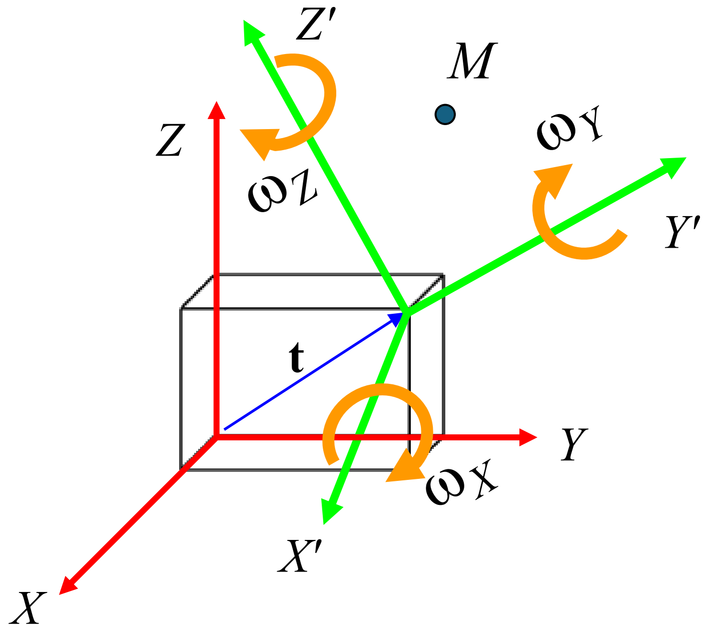
$$\mathbf{M} = [\, X \;\; Y \;\; Z \;\; 1 \,]^{\mathsf{T}}$$

[3D Transformation]



$$\mathbf{M} = [\ X\ \ Y\ \ Z\ \ 1\ ]^{\mathsf{T}}$$

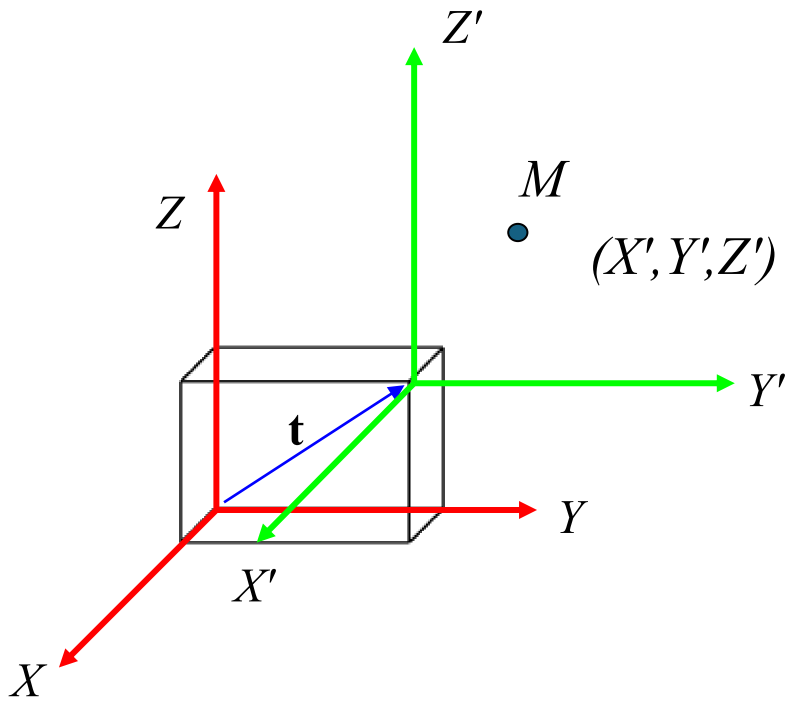
$$\mathbf{M'} = [\ X'\ \ Y'\ \ Z'\ \ 1\ ]^{\mathsf{T}}$$



$\mathbf{t}=(t_X, t_Y, t_Z)$: Translation

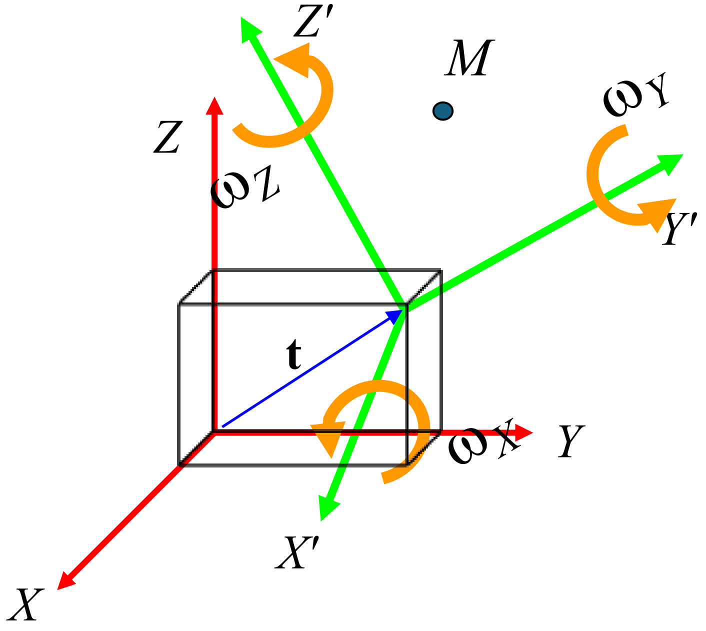
$\omega_X, \omega_Y, \omega_Z$: Rotation

3D Translation:

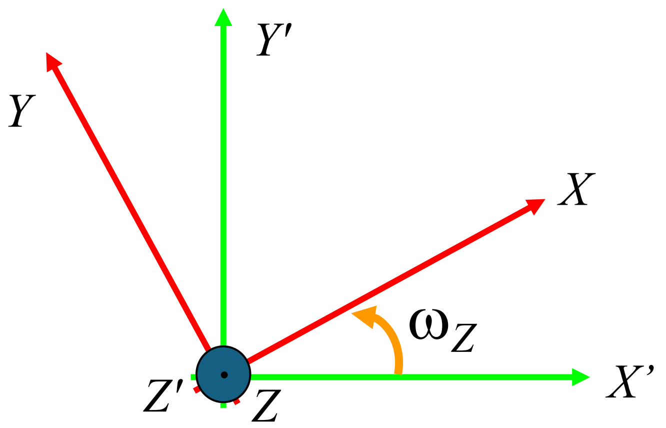


$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} X'+t_x \\ Y'+t_y \\ Z'+t_z \end{bmatrix} \longrightarrow \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & t_x \\ 0 & 1 & 0 & t_y \\ 0 & 0 & 1 & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

3D Rotation:

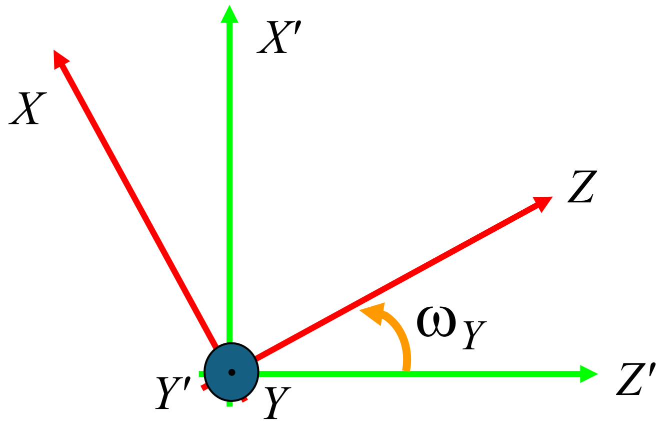


Axis Z Rotation:



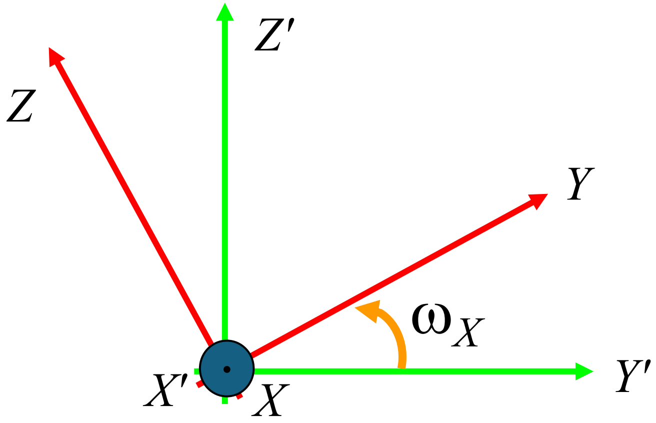
$$\mathbf{R}_Z = \begin{bmatrix} \cos(\omega_Z) & \sin(\omega_Z) & 0 \\ -\sin(\omega_Z) & \cos(\omega_Z) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Axis Y Rotation:

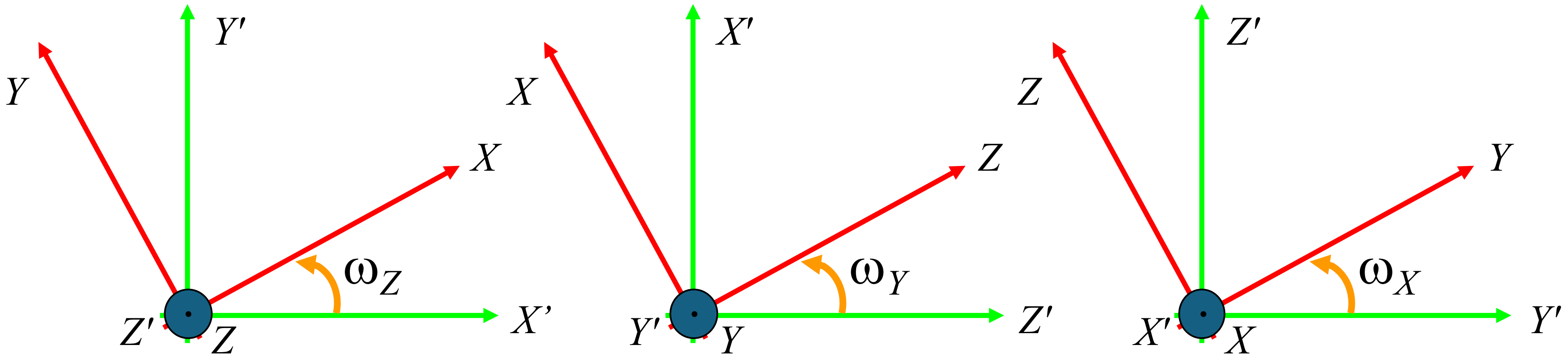


$$\mathbf{R}_\mathbf{Y} = \begin{bmatrix} \cos(\omega_Y) & 0 & -\sin(\omega_Y) \\ 0 & 1 & 0 \\ \sin(\omega_Y) & 0 & \cos(\omega_Y) \end{bmatrix}$$

Axis X Rotation :



$$\mathbf{R}_X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\omega_X) & \sin(\omega_X) \\ 0 & -\sin(\omega_X) & \cos(\omega_X) \end{bmatrix}$$



$$\mathbf{R}_Z = \begin{bmatrix} \cos(\omega_Z) & \sin(\omega_Z) & 0 \\ -\sin(\omega_Z) & \cos(\omega_Z) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_Y = \begin{bmatrix} \cos(\omega_Y) & 0 & -\sin(\omega_Y) \\ 0 & 1 & 0 \\ \sin(\omega_Y) & 0 & \cos(\omega_Y) \end{bmatrix}$$

$$\mathbf{R}_X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\omega_X) & \sin(\omega_X) \\ 0 & -\sin(\omega_X) & \cos(\omega_X) \end{bmatrix}$$

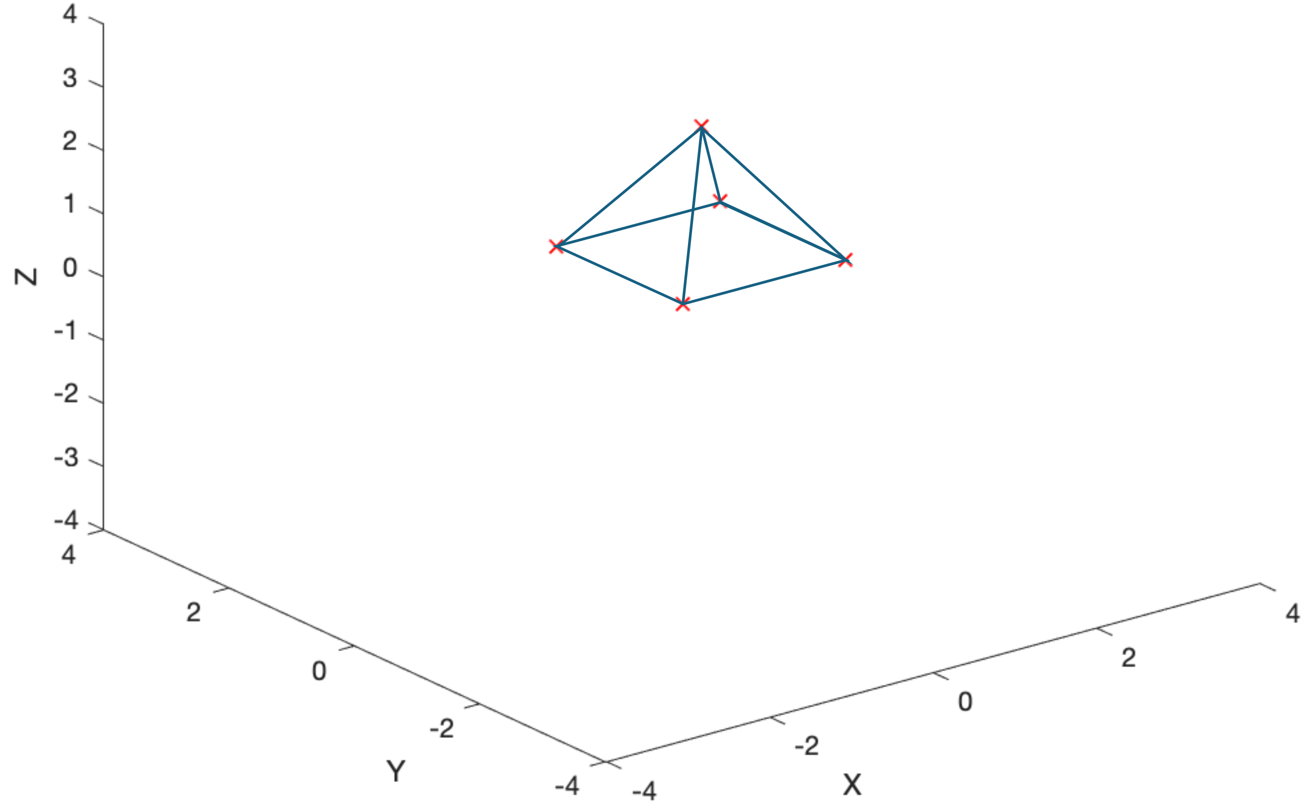
$$\mathbf{R}(\omega_X,\omega_Y,\omega_Z) = \mathbf{R}_X(\omega_X)\mathbf{R}_Y(\omega_Y)\mathbf{R}_Z(\omega_Z) = \begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix}$$

$$\begin{aligned}
R_{11} &= \cos(\omega_Y)\cos(\omega_Z) \\
R_{12} &= \cos(\omega_Y)\sin(\omega_Z) \\
R_{13} &= -\sin(\omega_Y) \\
R_{21} &= \sin(\omega_X)\sin(\omega_Y)\cos(\omega_Z) - \cos(\omega_X)\sin(\omega_Z) \\
R_{22} &= \sin(\omega_X)\sin(\omega_Y)\sin(\omega_Z) + \cos(\omega_X)\cos(\omega_Z) \quad . \\
R_{23} &= \sin(\omega_X)\cos(\omega_Y) \\
R_{31} &= \cos(\omega_X)\sin(\omega_Y)\cos(\omega_Z) + \sin(\omega_X)\sin(\omega_Z) \\
R_{32} &= \cos(\omega_X)\sin(\omega_Y)\sin(\omega_Z) - \sin(\omega_X)\cos(\omega_Z) \\
R_{33} &= \cos(\omega_X)\cos(\omega_Y)
\end{aligned}$$

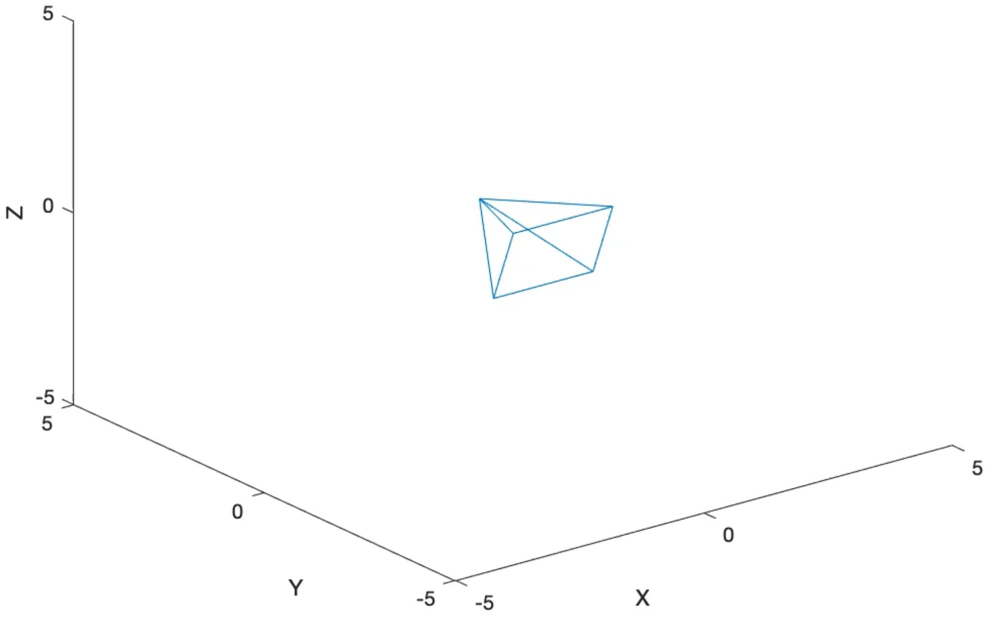
[3D Transformation]

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

[Example: Pyramid]

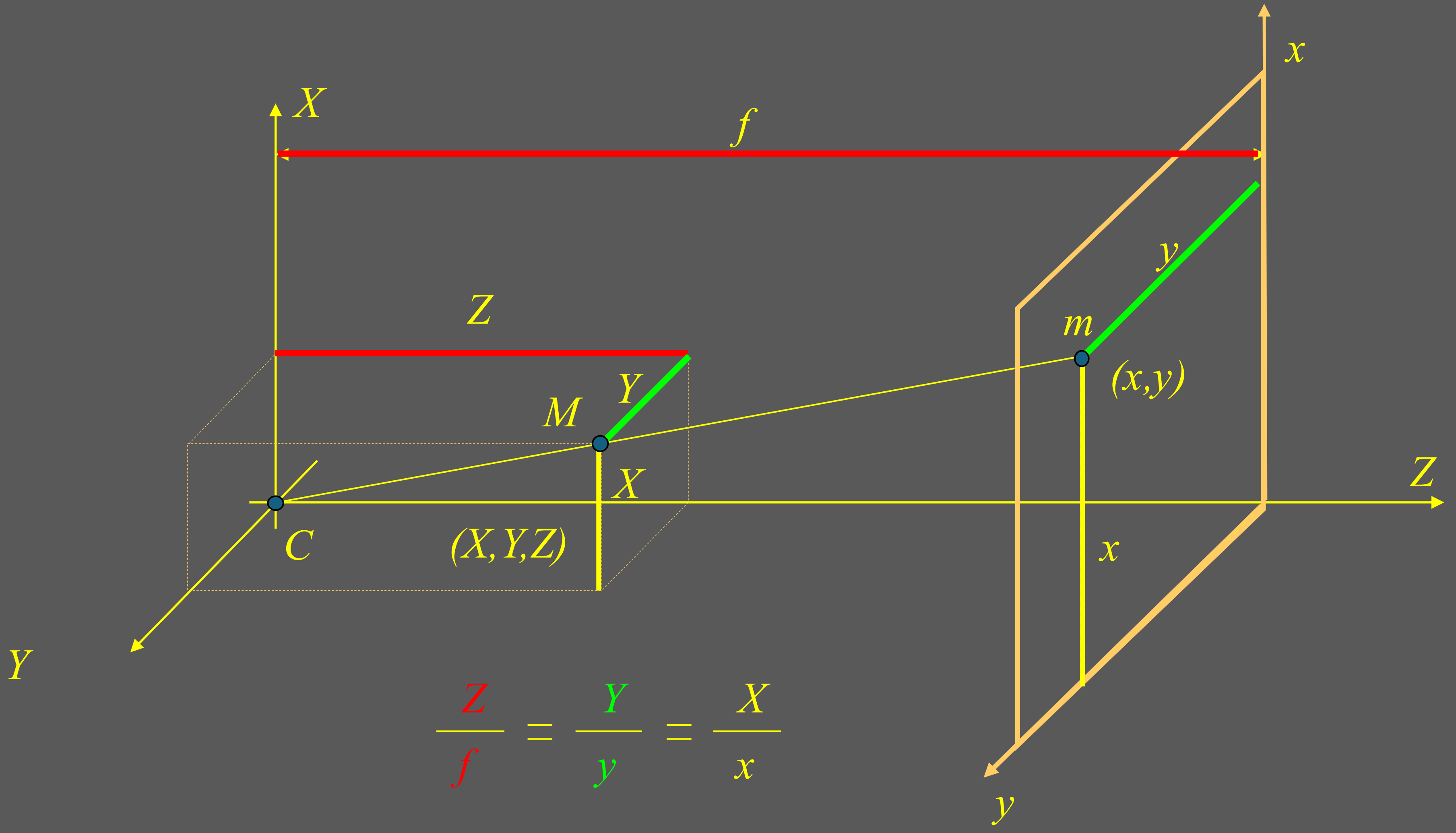


```
t = ( 0.00, 0.00, 0.00) w = ( 0.00, 0.00, 0.00)
t = ( 1.00, 2.00, 3.00) w = ( 0.00, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.57, 0.00, 0.00)
t = ( 0.00, 0.00, 2.00) w = ( 3.14, 0.00, 0.00)
t = ( 2.00, 2.00, 0.00) w = ( 0.00, 0.79,-0.39)
starting pyramid...
t = ( 0.00, 0.00, 0.00) w = ( 0.00, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.09, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.17, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.26, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.35, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.44, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.52, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.61, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.70, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.79, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.87, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 0.96, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.05, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.13, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.22, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.31, 0.00, 0.00)
t = ( 0.00, 0.00, 0.00) w = ( 1.40, 0.00, 0.00)
```



Transformaciones 3D > 2D

[Perspective Projection]



$$\frac{Z}{f} = \frac{Y}{y} = \frac{X}{x}$$

[Perspective Projection]

$$\frac{Z}{f} = \frac{Y}{y} = \frac{X}{x}$$

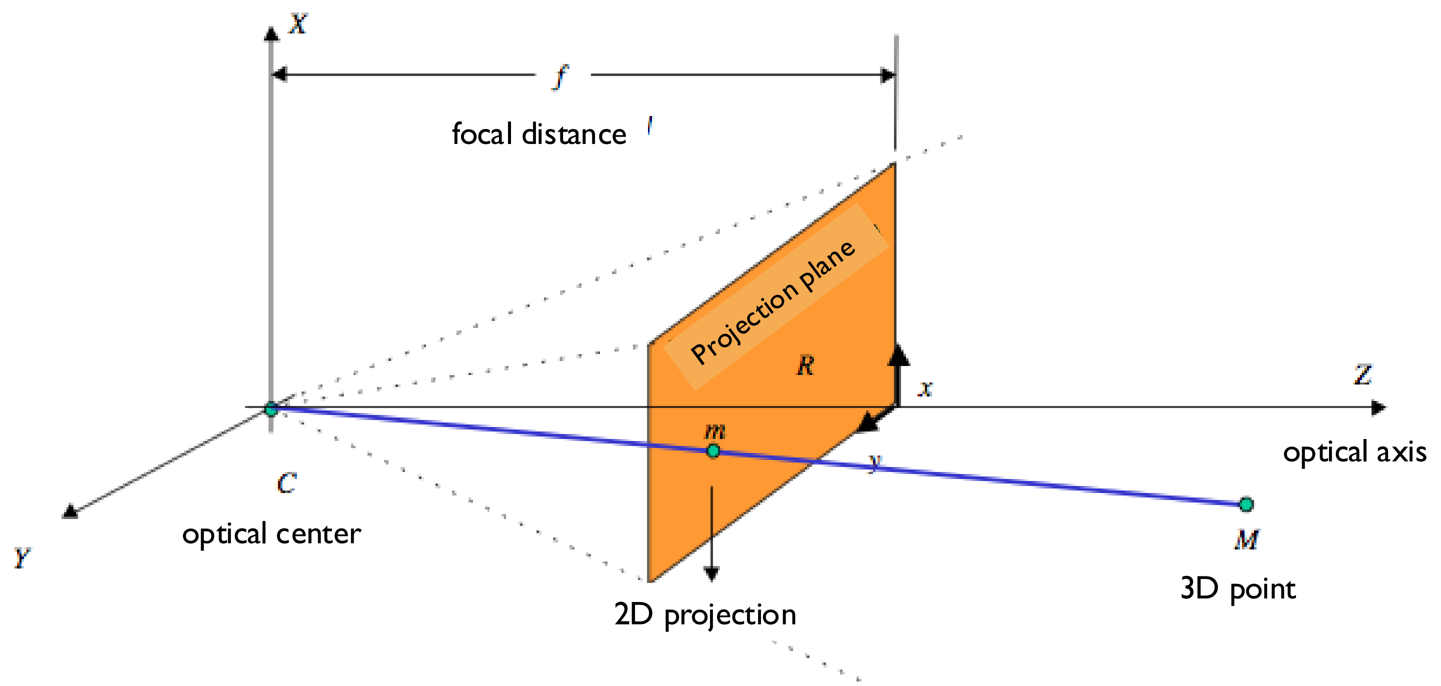
$$Z\,x = f\;X$$

$$Z\,y = f\;Y$$

$$Z\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} fX \\ fY \\ Z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

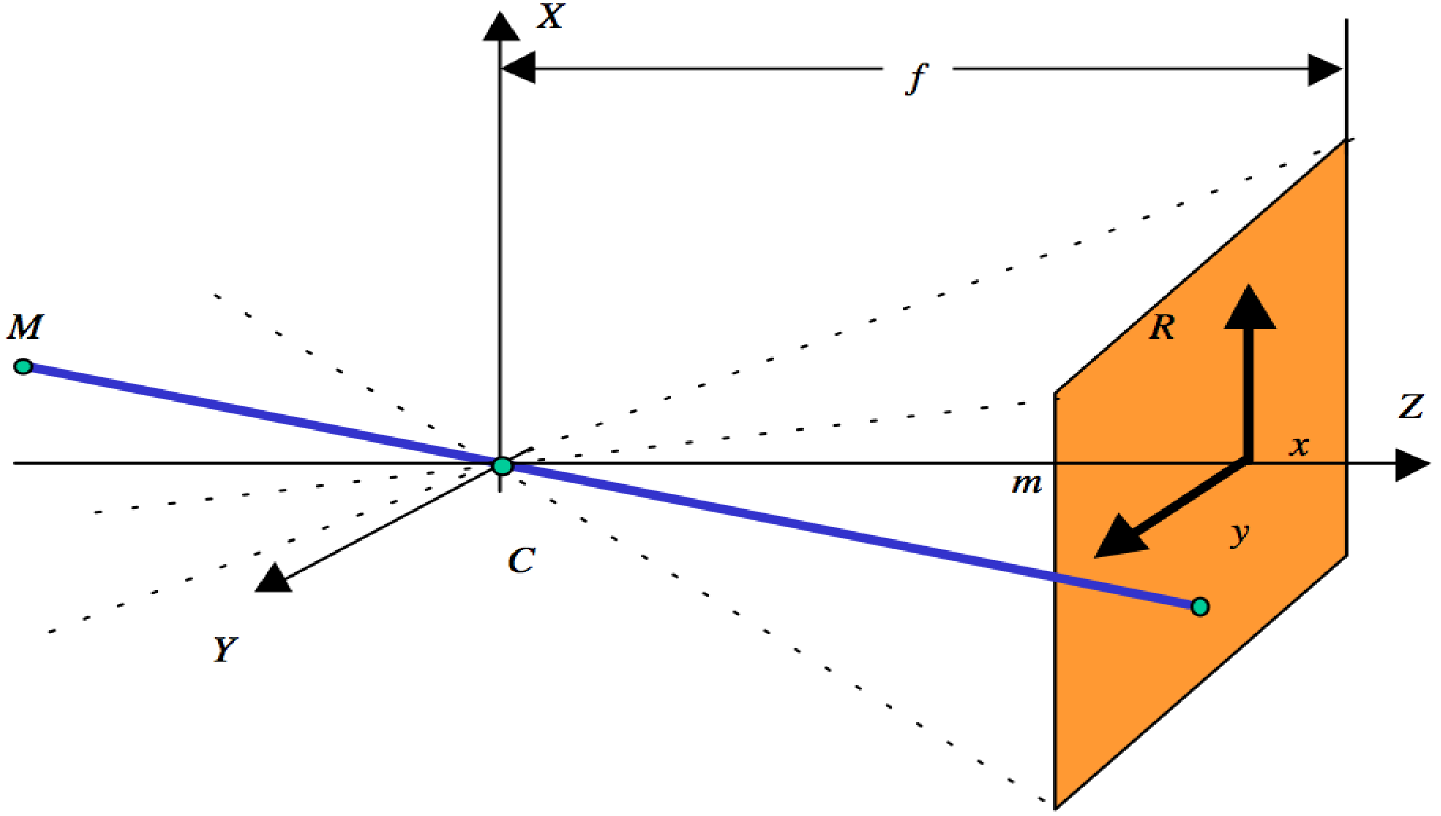
$\mathbf{m}$ is proportional to $\mathbf{P}$ * $\mathbf{M}$

$$\lambda\mathbf{m} = \mathbf{PM}$$



Matrix P = ?

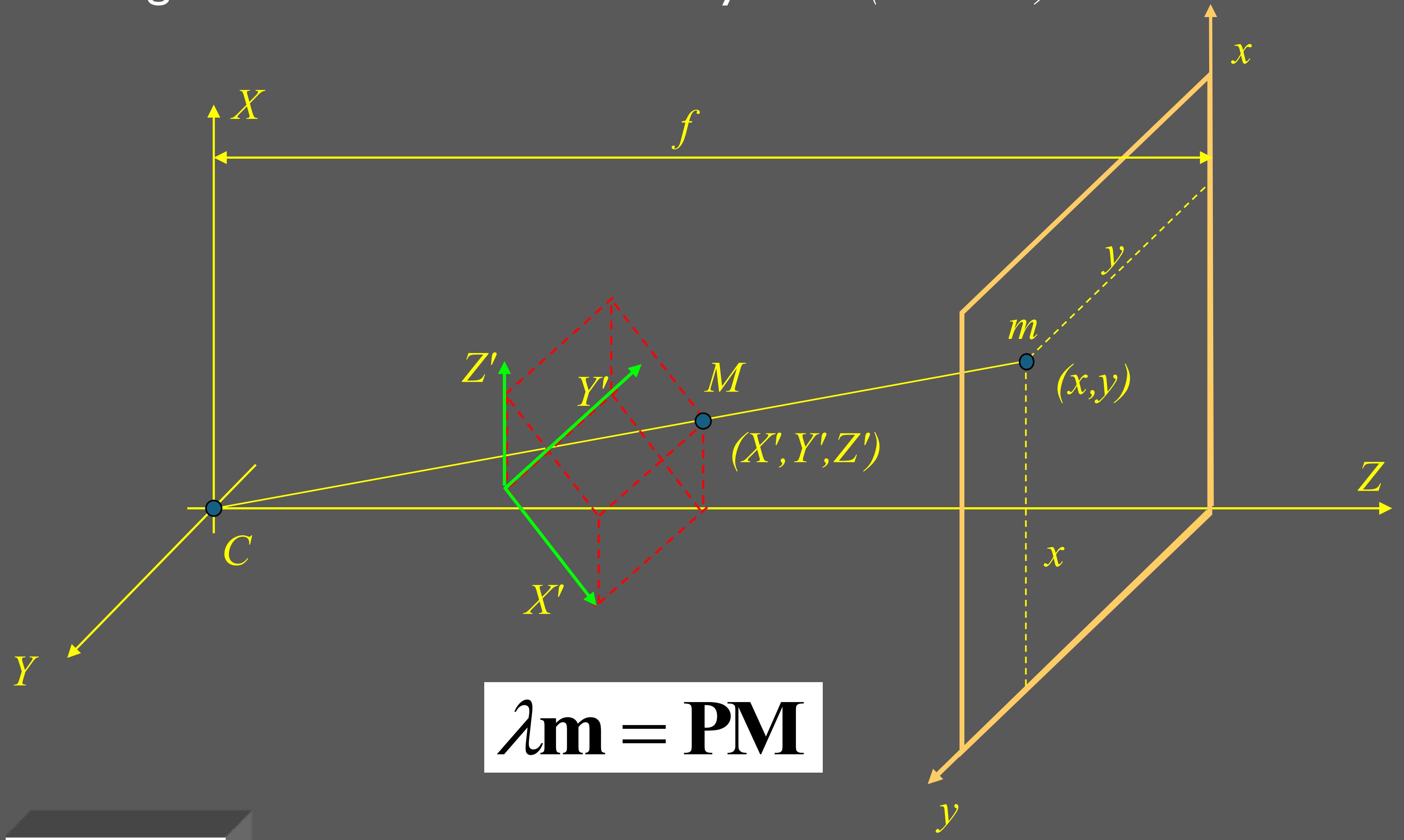
$$\begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



Matrix P = ?
$$\begin{bmatrix} -f & 0 & 0 & 0 \\ 0 & -f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

[Perspective Projection]

Now, M is given in a new coordinate system: (X',Y',Z')

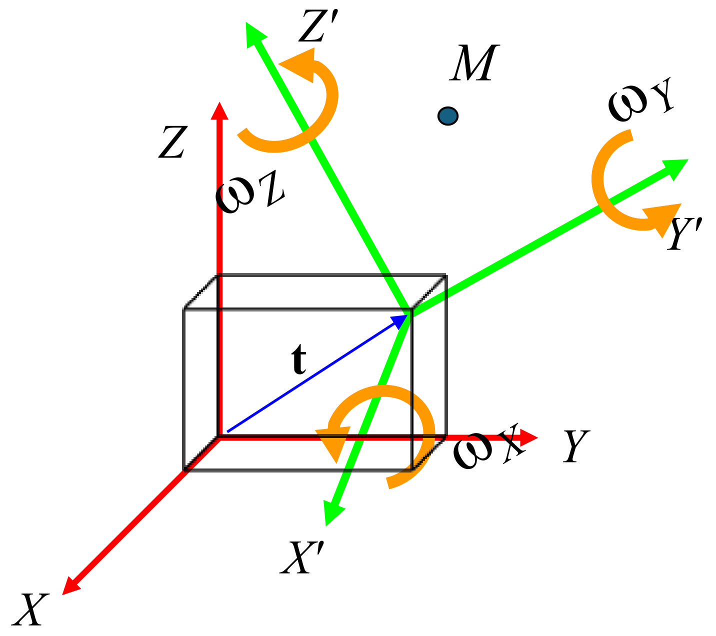


$$\lambda \mathbf{m} = \mathbf{PM}$$

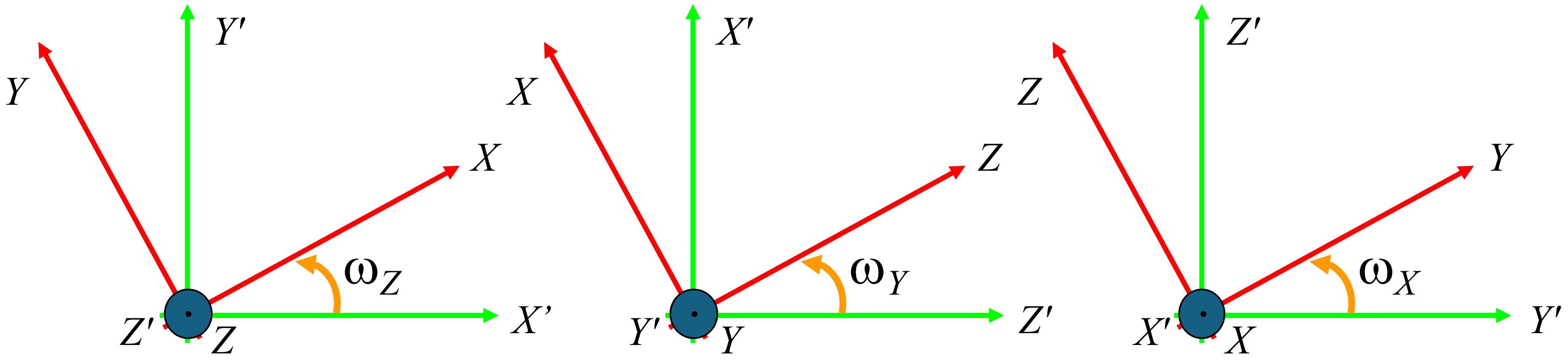
$$\lambda \mathrm{m} = \mathrm{PM}$$

However, $\mathbf{M}$ was given in (X,Y,Z) coordinate system

3D Rotation:



[3D Transformation]



$$\mathbf{R}_Z = \begin{bmatrix} \cos(\omega_Z) & \sin(\omega_Z) & 0 \\ -\sin(\omega_Z) & \cos(\omega_Z) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{R}_Y = \begin{bmatrix} \cos(\omega_Y) & 0 & -\sin(\omega_Y) \\ 0 & 1 & 0 \\ \sin(\omega_Y) & 0 & \cos(\omega_Y) \end{bmatrix}$$

$$\mathbf{R}_X = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\omega_X) & \sin(\omega_X) \\ 0 & -\sin(\omega_X) & \cos(\omega_X) \end{bmatrix}$$

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R_{22} &= \sin(\omega_X)\sin(\omega_Y)\sin(\omega_Z) + \cos(\omega_X)\cos(\omega_Z) \quad . \\
R_{23} &= \sin(\omega_X)\cos(\omega_Y) \\
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\end{aligned}$$

[3D Transformation]

$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0}^\top & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

[Perspective Projection]

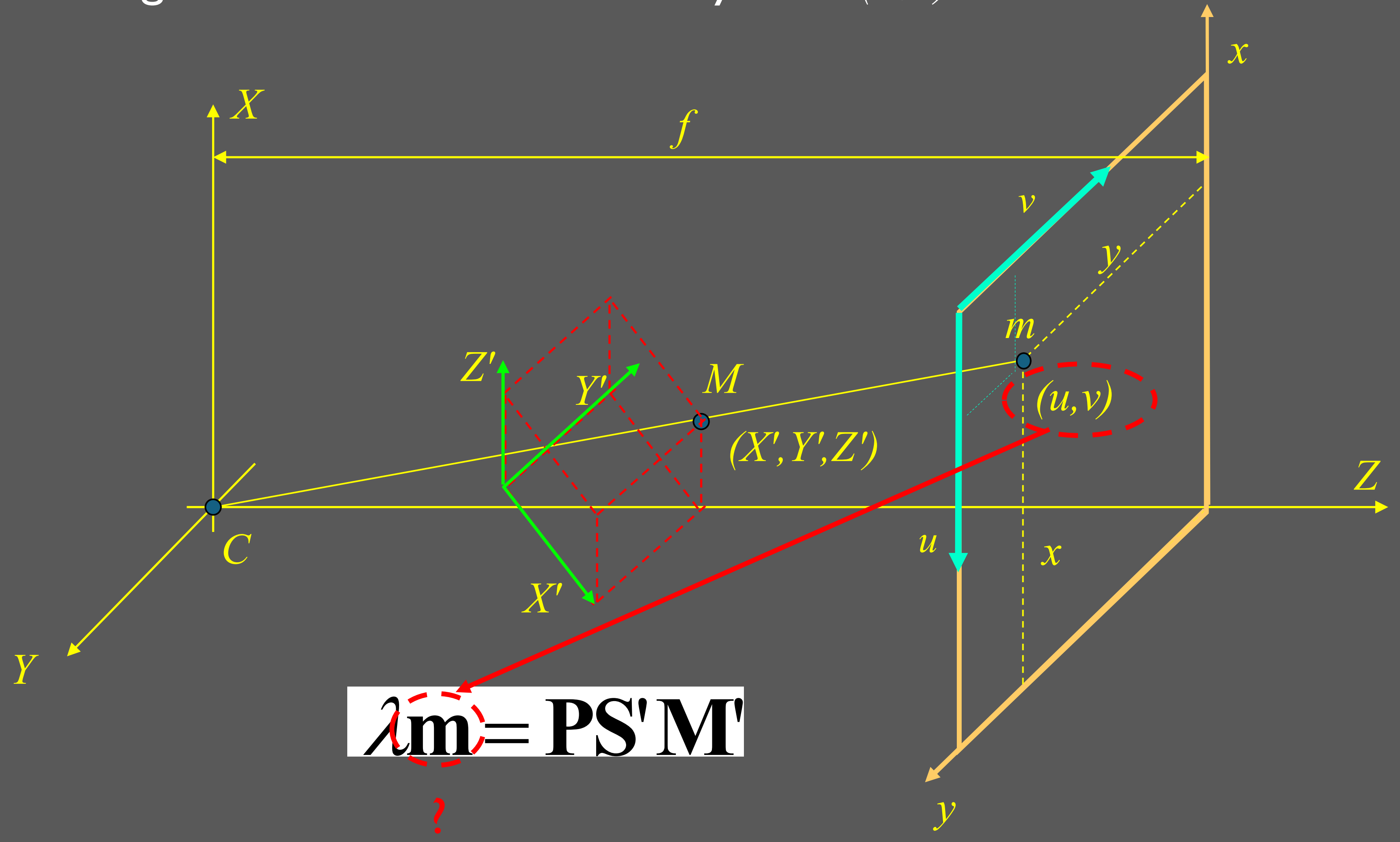
$$\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \mathrm{R}' & \mathbf{t}' \\ \mathbf{0}^{\mathsf{T}} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix} \longrightarrow \mathbf{M} = \underbrace{\mathbf{S}'\mathbf{M}'}$$

$$\lambda \mathbf{m} = \mathbf{P}\mathbf{M}$$

$$\lambda \mathbf{m} = \mathbf{P}\overbrace{\mathbf{S}'\mathbf{M}'}$$

[Perspective Projection]

Now, m is given in a new coordinate system: (u,v)



[Perspective Projection]

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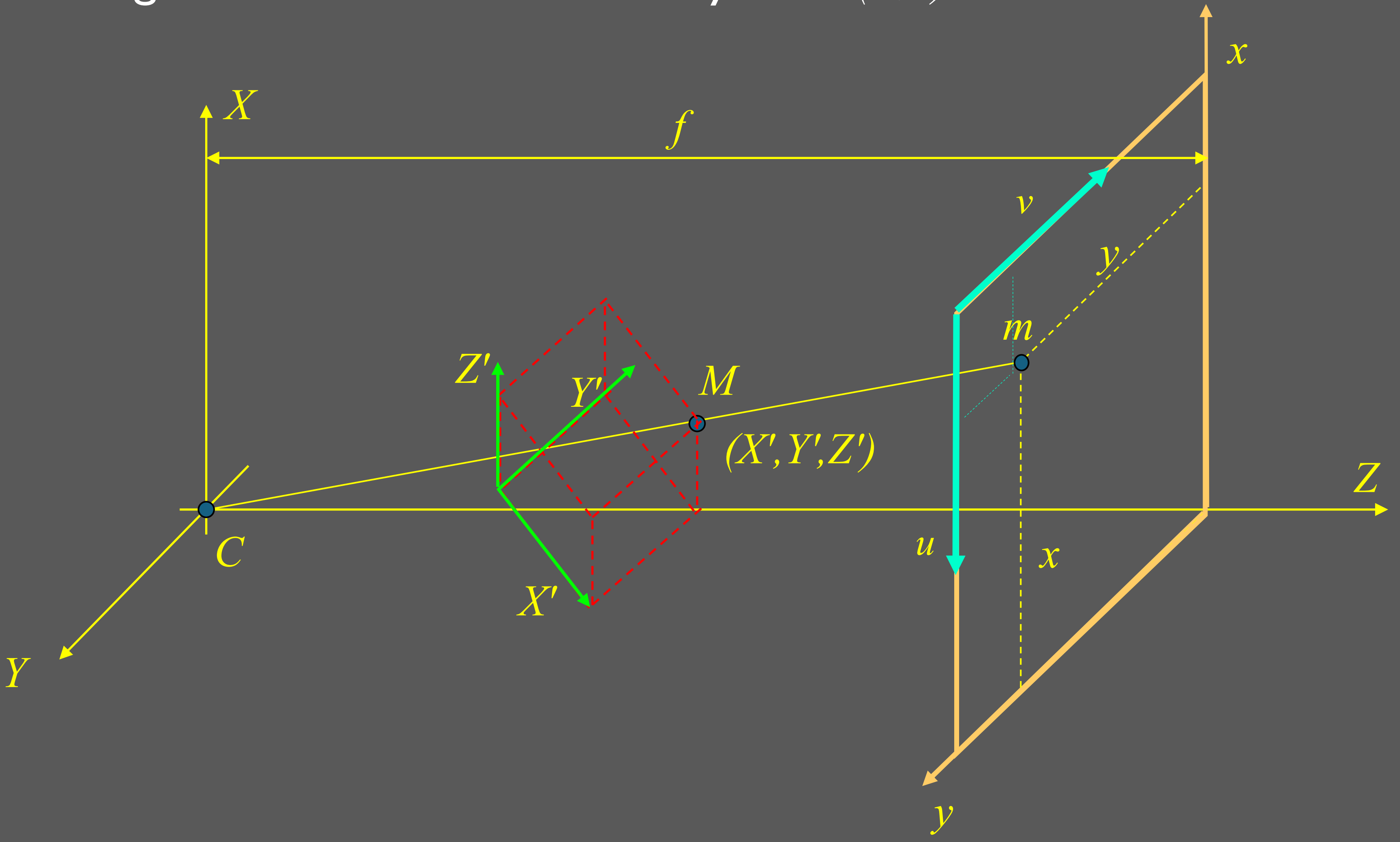

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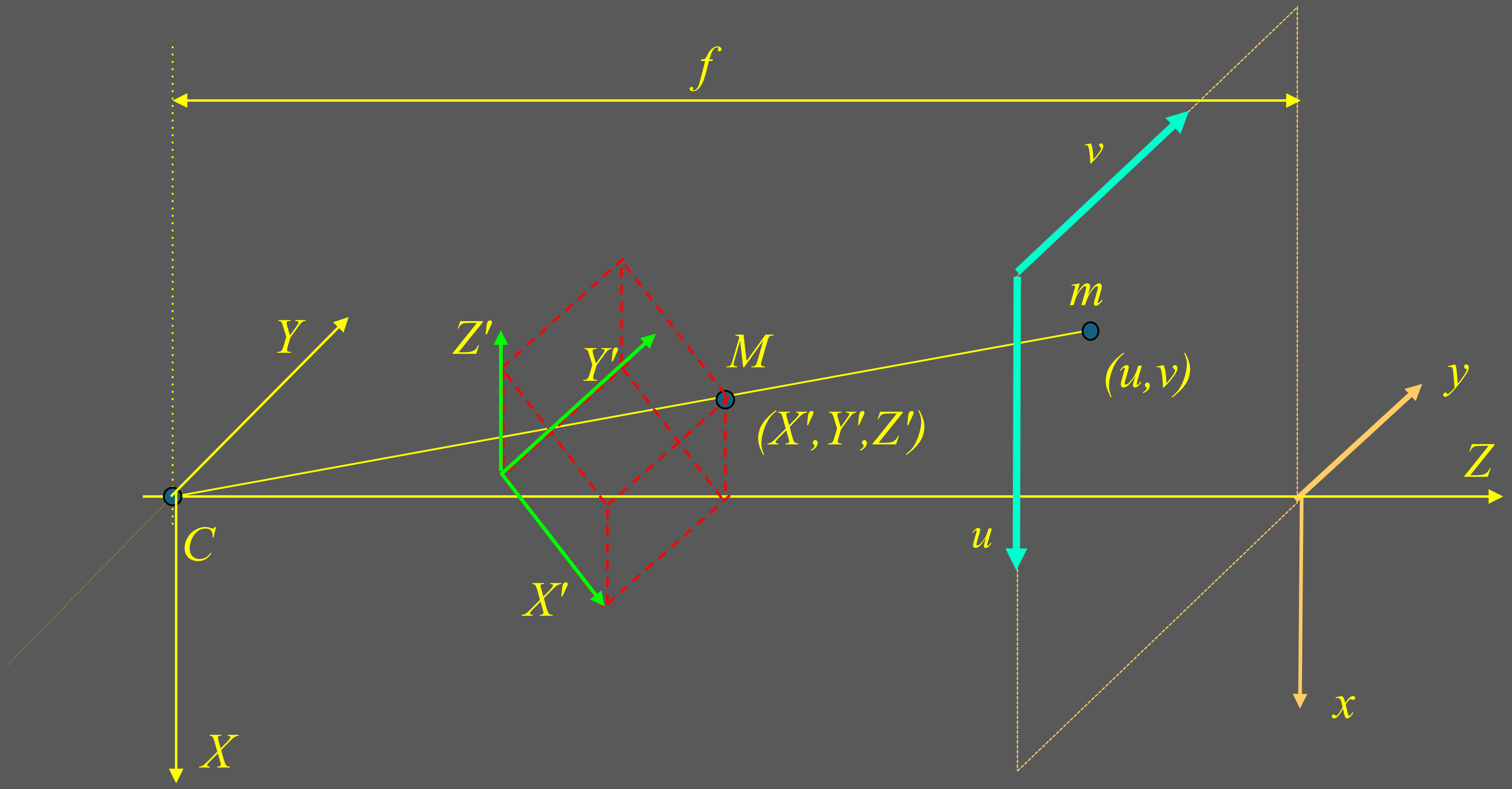
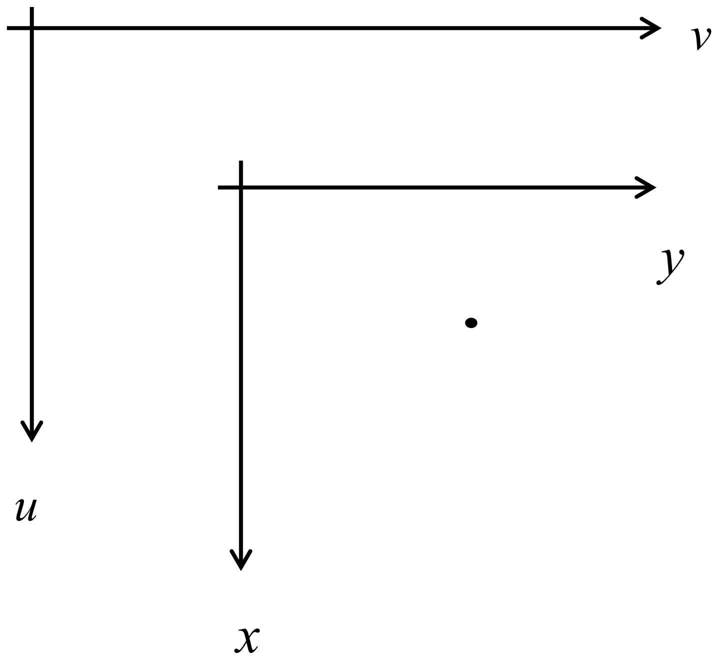
Now, m is given in a new coordinate system: (u,v)



[Perspective Projection]



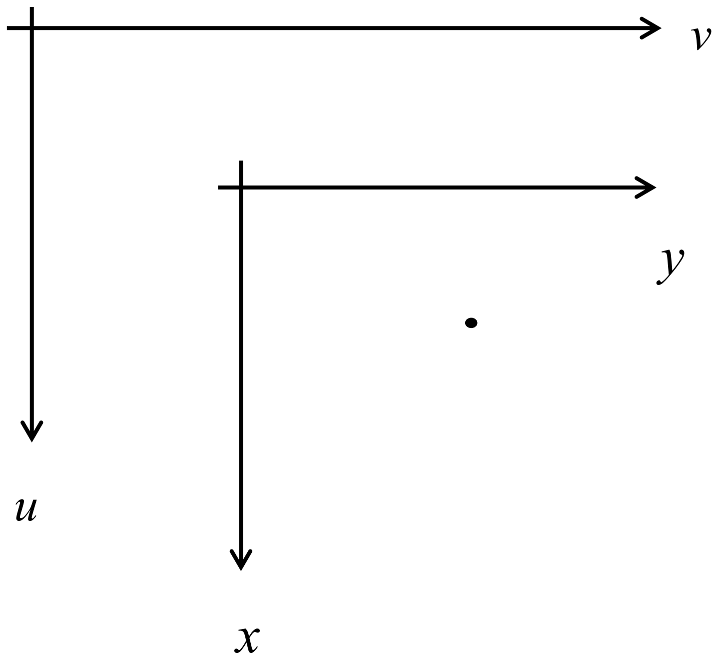
Between (x,y) and (u,v) there is no rotation.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & u_0 \\ 0 & 1 & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Is this OK?

No, because in a camera system (x,y) is given in mm or inches and (u,v) in pixels.

[Perspective Projection]



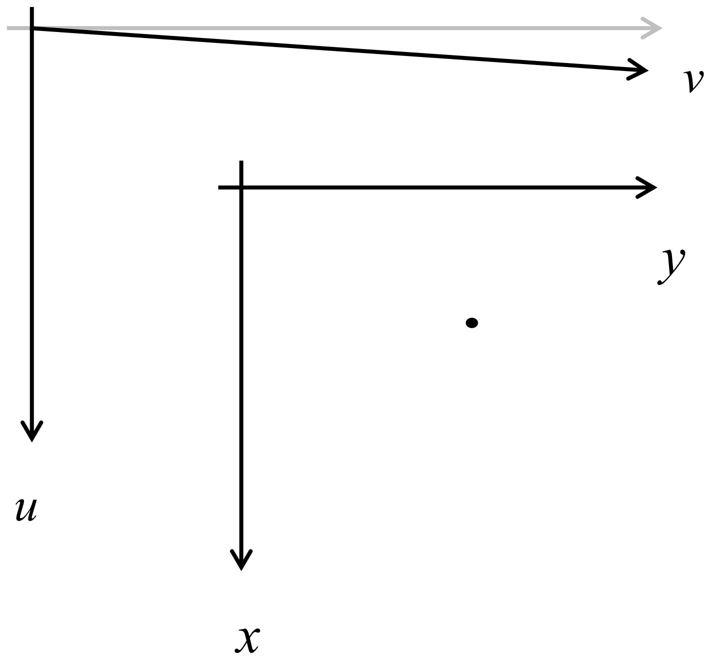
Between (x,y) and (u,v) there is no rotation.
There is a scale factor in each direction.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Is this OK?

No, because sometimes the angle between u and v is not 90, it could be for example 89.99.

[Perspective Projection]



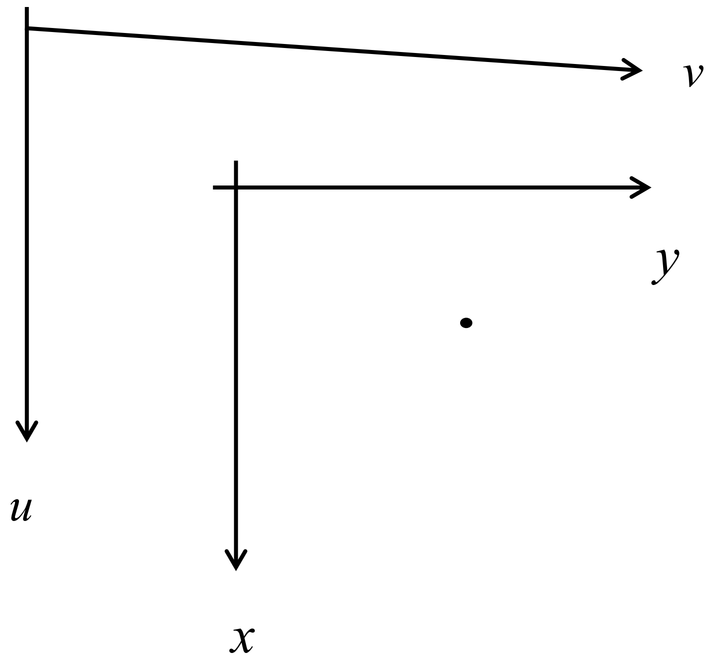
Between (x,y) and (u,v) there is no rotation.
There is a scale factor in each direction.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Is this OK?

No, because sometimes the angle between u and v is not 90, it could be for example 89.99.

[Perspective Projection]



Between (x,y) and (u,v) there is no rotation.
There is a scale factor in each direction.
There is a skew factor.

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Is this OK? Yes!

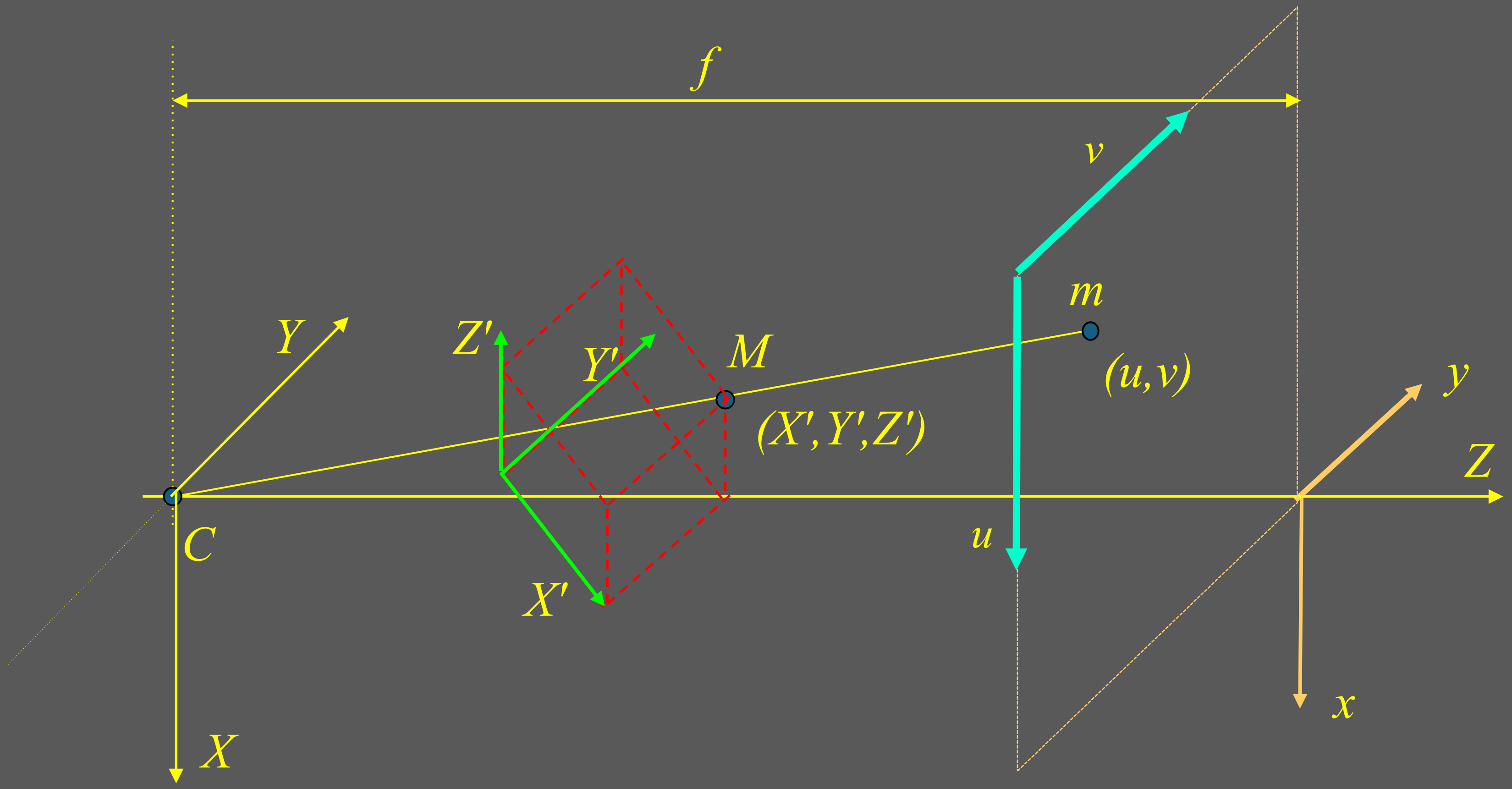
$$\mathbf{m} = [x \;\; y \;\; 1]^\mathsf{T}$$

$$\mathbf{w} = [u \;\; v \;\; 1]^\mathsf{T}$$

$$\mathbf{w} = \mathbf{K}\mathbf{m}$$

[Perspective Projection]

Now, m is given in a new coordinate system: (u,v)



$$\lambda \mathbf{w} = \mathbf{K P S' M'}.$$

[Camera Model]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f\alpha_x & fs & u_0 & 0 \\ 0 & f\alpha_y & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

[Camera Model]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

[Camera Model]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

11 DoF:

- 5 intrinsec parameters $(\alpha, \beta, \gamma, u_0, v_0)$
- 6 extrinsec parameters $(\omega_X, \omega_Y, \omega_Z), (t_X, t_Y, t_Z)$